

# Luis A. Saenz

MS CS @ UTSA | Systems Security | San Antonio, TX  
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## EDUCATION

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- 2027      **M.S., Computer Science**  
The University of Texas at San Antonio  
GPA: 3.94  
Relevant Coursework: Operating Systems, Systems Programming, Software Reverse Engineering, Secure Software Development, Computer Architecture, Programming Languages and Compilers
- 2025      **B.S., Computer Science**  
The University of Texas at San Antonio, *summa cum laude*  
GPA: 3.98  
Minor and concentration in Cybersecurity

## RESEARCH INTERESTS

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Systems security; memory corruption and heap security; program and binary analysis; exploit development; reverse engineering.

## RESEARCH EXPERIENCE

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- 2026 – 2027      **Master's Thesis: Dynamic Heap-State Analysis for Use-After-Free Exploitability**  
Advisor: Dr. Xiaoyin Wang, The University of Texas at San Antonio  
Developing a low perturbation, glibc aware runtime analysis framework for Linux x86-64 programs that reconstructs heap allocation generations and same address reuse, associates stale accesses with reclaimed objects, tracks attacker derived data in replacement allocations, and classifies the observed consequences of use-after-free executions.
- 2026      **Summer Graduate Research Assistant**  
The University of Texas at San Antonio  
Developed technical activities and provided implementation guidance for seven undergraduate teams working with nonprofit partners, including Goodwill and United Way. Supported symposium reports and posters, receiving formal acknowledgment for project and whitepaper contributions.

## TECHNICAL PROJECTS

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- 2026      **cage: Lightweight Linux Confinement Runtime**  
C, Linux, ptrace, namespaces, cgroups v2, seccomp-BPF.  
Developed a security oriented Linux sandbox for executing untrusted binaries using user, PID, mount, UTS, and network namespaces, with an isolated root filesystem created through bind mounts, pivot-root, and a private /proc. Implemented ptrace-based syscall monitoring, configurable seccomp-BPF allowlists, and cgroups v2 controls to restrict syscall access and resource consumption within the sandbox.
- 2026      **Decaf Compiler Construction Project**  
C++, Flex, Bison, MIPS  
Implemented lexical analysis, parsing, semantic analysis, type checking, and MIPS code generation for a Decaf language compiler.
- 2024 – 2026      **LuRETools: Binary Analysis Toolkit**  
C, ELF, PE  
Built a modular C toolkit that parses ELF and PE binaries into a common analysis model containing architecture, entry point, section permissions, symbols, and function metadata. Implemented command line workflows for binary inspection and intermediate representation listing, enabling consistent analysis across executable formats.
- 2025      **LLInj: Raw Packet Injection Tool**  
C, libpcap, libdnet, Ethernet, TCP/IP  
Built a configurable tool that parses PCAP captures, decodes Ethernet, ARP, IPv4, TCP, and UDP headers, and injects selected packets through a network interface. Rewrote MAC addresses, IP addresses, and transport layer ports for new endpoints, recalculated protocol checksums, and tracked TCP sequence and acknowledgment state from captured responses.
- 2025      **WannaCry Analysis Case Study**  
FLARE VM, Ghidra, WinDbg  
Reverse engineered WannaCry's EternalBlue-based SMBv1 exploit chain and analyzed its AES/RSA encryption workflow, mutex based execution controls, and lateral movement behavior through Windows service scanning.

## TEACHING EXPERIENCE

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- 2025 – 2026      **Graduate Grader**  
The University of Texas at San Antonio  
Evaluated programming assignments and provided feedback on software architecture, design patterns, and code quality.

## TECHNICAL SKILLS

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**Programming:** C/C++, Python, Java, x86/x86-64, ARM64

**Systems Security:** Reverse engineering, binary analysis, exploit development, memory-corruption debugging, coverage-guided fuzzing

**Systems Programming:** Linux internals, processes, signals, system calls, pthreads, synchronization, IPC, sockets, namespaces, cgroups v2, seccomp-BPF

**Tools:** GDB, Ghidra, WinDbg, pwntools, AFL++, libFuzzer, ASan/UBSan, strace, QEMU

**Analysis Environments:** REMnux, FLARE VM, Proxmox, Docker

## AWARDS AND HONORS

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2023 – 2025     Digital Defense Endowment Scholarship, Digital Defense Inc.

2023 – 2025     CNF Cyber Security Scholarship, CNF Technologies

## TECHNICAL OUTREACH AND TRAINING

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2025 – present     **Reverse Engineering and Systems Security YouTube: Luis' Low Level**  
Technical videos on malware analysis, reverse engineering, systems security, and exploit development.

2025 – present     **pwn.college Binary Exploitation and Program Security Training**  
Completed challenges in Linux exploitation, shellcoding, memory corruption, ROP, sandbox escaping, and syscall level security.

## ADDITIONAL PROFESSIONAL EXPERIENCE

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2025 – 2026     **Founder and Lead Engineer, erratumAI**  
Built a full-stack AI-powered textbook correction platform using Spring Boot, React, PostgreSQL, Redis and GCP.